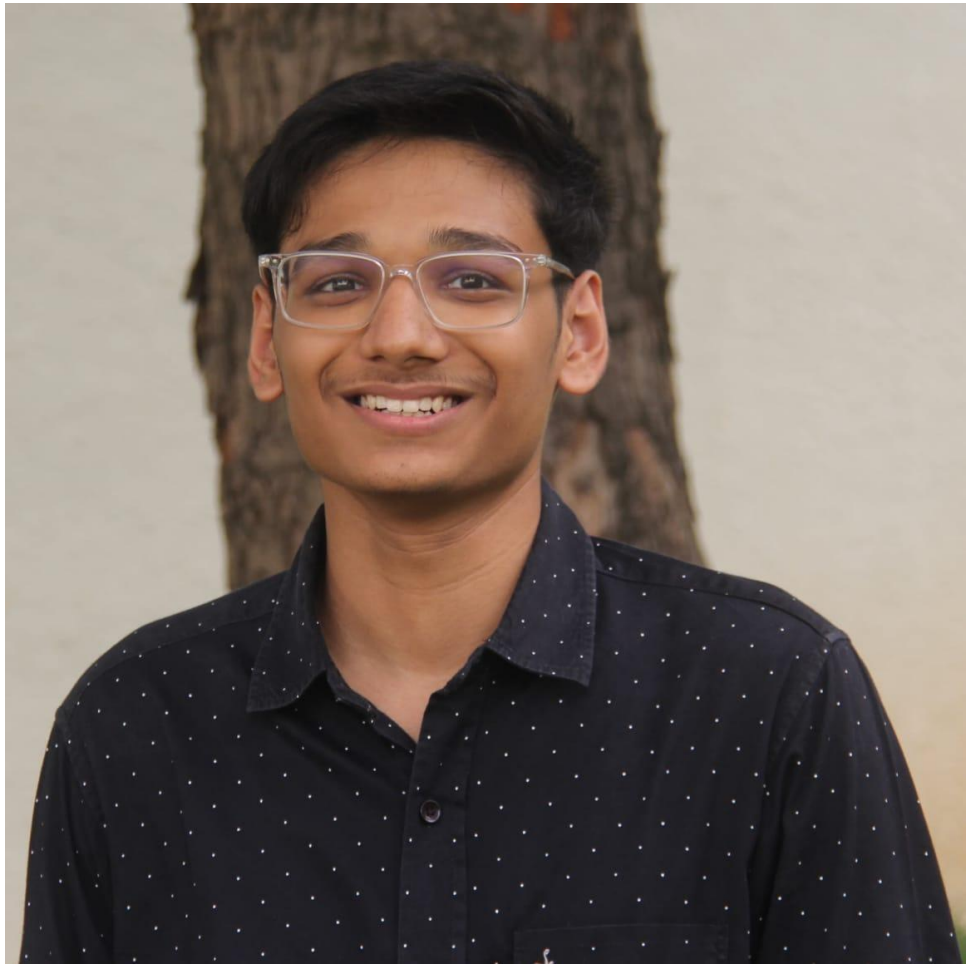
Microsoft Elevate

CAPSTONE PROJECT

Air Quality Intelligence:

Visualizing Pollution Trends in Indian Cities



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PROBLEM STATEMENT:

Air pollution has become one of the most critical environmental and public health challenges affecting urban populations worldwide, contributing to respiratory illnesses, reduced quality of life, and increased mortality rates. Cities continuously generate large volumes of air quality data, including AQI levels and pollutant concentrations such as PM2.5, PM10, and NO_2, but this data often remains difficult to interpret due to its scale and complexity. As a result, identifying pollution hotspots, understanding how air quality changes over time, and determining which pollutants are most responsible for deteriorating conditions becomes challenging for policymakers, researchers, and the public. Without clear and accessible analytical tools, this valuable data cannot effectively support environmental awareness or data-driven decision-making.

PROPOSED SOLUTION:

The proposed system aims to address the challenge of understanding and analyzing urban air pollution patterns by transforming complex air quality datasets into clear and interactive visual insights. This involves using data analytics and visualization techniques to identify pollution hotspots, analyze AQI trends over time, and understand the contribution of major pollutants affecting air quality. The solution will consist of the following components:

- **Data Collection:**
  Gather historical air quality data from reliable sources, including AQI values and pollutant concentrations such as PM2.5, PM10, and NO_2 across multiple cities and time periods. The dataset includes city-level pollution measurements and categorical AQI levels that indicate air quality severity.

- **Data Preprocessing:**
  Clean and preprocess the collected data to handle missing values, remove inconsistencies, and ensure proper data formatting. Extract relevant attributes such as year and pollutant averages to enable meaningful analysis and comparison across cities and time periods.

- **Machine Learning Component:**
  An ML model can be incorporated to analyze historical pollution patterns and predict future AQI trends. Time-series forecasting methods such as ARIMA, Linear Regression, or LSTM can be used to estimate future air quality levels based on historical pollutant concentrations and temporal patterns. This helps identify potential future pollution risks and understand how different pollutants influence AQI levels.

PROPOSED SOLUTION:

- **Data Analysis and Visualization:**
  Use analytical techniques to identify patterns in air quality data and present them through intuitive visualizations. Key insights include identifying the most polluted cities, analyzing the distribution of AQI categories, observing how air quality changes over time, and comparing pollutant concentrations to determine their impact on overall air quality.

- **Interactive Dashboard Development:**
  Develop an interactive Power BI dashboard that allows users to explore the data dynamically through visual elements such as KPI indicators, bar charts, pie charts, trend lines, and pollutant comparison charts. Filters and slicers enable users to analyze air quality by specific cities and years.

- **Evaluation and Insight Generation:**
  Evaluate the effectiveness of the dashboard by ensuring the visualizations clearly communicate pollution patterns and trends. The system enables users to easily interpret complex environmental data, supporting awareness and facilitating data-driven discussions regarding urban air pollution.

SYSTEM APPROACH:

- The "**System Approach**" section outlines the overall strategy and methodology for developing and implementing the Air Quality Intelligence Dashboard. The system focuses on collecting air quality data, preprocessing it for analysis, and presenting meaningful insights through an interactive Power BI dashboard that allows users to explore pollution patterns across cities and time periods.

System Requirements-

- The system requires a computing environment capable of handling data analysis and visualization tasks. This includes a computer with sufficient processing power and memory to manage datasets and run Power BI smoothly. The software requirements include Microsoft Power BI Desktop for dashboard development and data visualization, along with spreadsheet tools such as Microsoft Excel or CSV-compatible editors for preliminary data inspection and preparation. The system also requires access to the air quality dataset containing AQI values and pollutant concentrations such as PM2.5, PM10, and NO_2.

SYSTEM APPROACH:

Libraries / Tools Required to Build the Model-

- The project primarily uses **Power BI** as the main visualization and analytics platform. Power BI provides built-in data transformation tools (Power Query) for data cleaning and preprocessing, as well as DAX (Data Analysis Expressions) for calculating key metrics such as average AQI, maximum AQI, and other aggregated values. Additional tools such as Microsoft Excel or CSV data formats are used for storing and importing the dataset. Power BI's visualization capabilities, including bar charts, line charts, pie charts, KPI cards, and slicers, are utilized to create an interactive dashboard that enables dynamic exploration of air quality data across cities and years.

ALGORITHM & DEPLOYMENT:

- **Algorithm Selection**

For this project, a **time-series forecasting approach using Linear Regression or ARIMA** can be used to analyze historical air quality data and predict future AQI trends. These algorithms are suitable because air pollution data is sequential and time-dependent, meaning past pollution levels can influence future values. Time-series models help capture patterns, seasonal variations, and long-term pollution trends across cities.

- **Data Input**

The algorithm uses historical air quality data as input. Key input features include **AQI values, pollutant concentrations (PM2.5, PM10, and NO_2), city information, and time-related attributes such as year or date**. These features help the model learn how pollutant levels and time patterns influence overall air quality.

ALGORITHM & DEPLOYMENT:

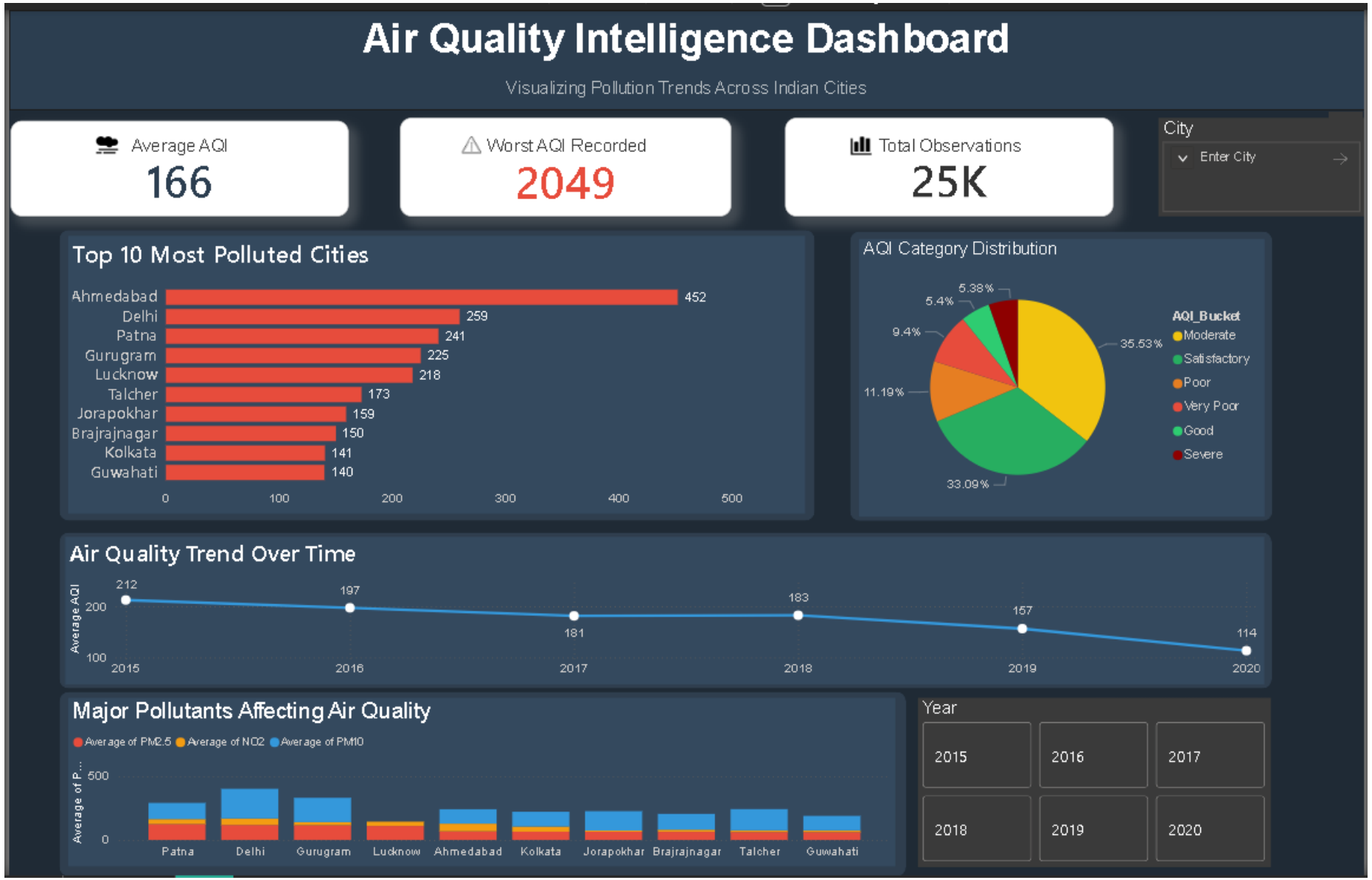
- **Training Process**

The model is trained using historical pollution records collected over multiple years. During training, the algorithm learns relationships between pollutant concentrations and AQI values. Data preprocessing techniques such as cleaning missing values and aggregating pollutant averages are applied before training. Model performance can be improved through parameter tuning and validation using a portion of the dataset for testing.

- **Prediction Process**

After training, the model can analyze new or recent pollution data to estimate future AQI trends. Based on learned patterns, the algorithm predicts how air quality may change in upcoming periods. These predictions can help identify potential pollution risks and provide additional insights alongside the visual analytics presented in the dashboard.

RESULT:


Air Quality Intelligence Dashboard
Visualizing Pollution Trends Across Indian Cities
Average AQI
166
Worst AQI Recorded
2049
Total Observations
25K
City
Enter City
Top 10 Most Polluted Cities
Ahmedabad 452
Delhi 259
Patna 241
Gurugram 225
Lucknow 218
Talcher 173
Jorapokhar 159
Brajrajnagar 150
Kolkata 141
Guwahati 140
0 100 200 300 400 500
AQI Category Distribution
5.38%
5.4%
9.4%
35.53%
11.19%
33.09%
AQI_Bucket
Moderate
Satisfactory
Poor
Very Poor
Good
Severe
Air Quality Trend Over Time
Average AQI
212 197 181 183 157 114
2015 2016 2017 2018 2019 2020
Major Pollutants Affecting Air Quality
Average of PM2.5 Average of NO2 Average of PM10
Average of P...
500
0
Patna Delhi Gurugram Lucknow Ahmedabad Kolkata Jorapokhar Brajrajnagar Talcher Guwahati
Year
2015 2016 2017
2018 2019 2020

CONCLUSION:

This project demonstrates how data analytics and visualization can transform complex environmental data into meaningful insights for understanding urban air pollution. By developing the **Air Quality Intelligence Dashboard**, the system effectively highlights pollution hotspots, reveals AQI trends over time, and identifies the major pollutants contributing to deteriorating air quality across cities. The interactive visualizations enable users to quickly interpret large volumes of air quality data that would otherwise be difficult to analyze in raw form. During development, challenges such as handling incomplete data, designing intuitive visualizations, and ensuring meaningful comparisons across cities were addressed through careful data preprocessing and dashboard structuring. In the future, the system can be enhanced by integrating real-time air quality monitoring and more advanced predictive models to forecast pollution levels. Ultimately, this project emphasizes the critical role of data-driven insights in increasing environmental awareness and supporting informed decision-making to address the growing challenge of urban air pollution.

FUTURE SCOPE:

The Air Quality Intelligence Dashboard can be expanded into a more powerful decision-support platform by integrating real-time air quality monitoring data from IoT sensors, government pollution monitoring stations, and satellite-based environmental datasets. Incorporating advanced machine learning and deep learning models can enable more accurate forecasting of pollution levels, allowing early detection of potential air quality deterioration and supporting proactive environmental planning. The system can also be extended to include additional influencing factors such as weather conditions, traffic density, industrial emissions, and seasonal variations to provide deeper insights into the causes of urban air pollution. Furthermore, the dashboard can be scaled to monitor multiple regions or countries, enabling large-scale comparative analysis of air quality trends. In the long term, integrating smart city technologies, real-time alerts, and predictive environmental analytics could help policymakers, researchers, and urban planners make data-driven decisions to reduce pollution levels and improve public health outcomes.

REFERENCES:

- **Dataset:** Air Quality Data in India (2015–2020), sourced from Kaggle.com. This dataset was used for analyzing pollution trends, AQI levels, and pollutant concentrations across multiple Indian cities.
  Link: https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india

- **Power BI Documentation:** Official Microsoft documentation used for understanding data visualization techniques, dashboard development, and analytical features in Power BI.
  Link: https://learn.microsoft.com/en-us/power-bi/

GitHub Link: https://github.com/RishabhGupta2005/air-quality-intelligence-dashboard

Thank You